

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as Intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I S.V.DILP (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

S.V. 

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1) Delivery Drone Problem. Solving Agent : Problem understanding.

An e-commerce company want to use an autonomous drone to deliver package within a city the drone has to find a safe and efficient route from its starting location to the customer's destination while avoiding obstacles.

Method Selection.

A Goal-Based Problem-Solving Agent is suitable because the drone has a specific goal: Successfully deliver the package to the destination.

the drone can use the A* Search Algorithm to select an efficient route.

$$f(n) = g(n) + h(n)$$

where,

*) $g(n)$ = actual travelling cost

*) $h(n)$ = estimated cost to destination.

*) $f(n)$ = total estimated cost.

Solution Process / pseudocode.

- 1) Set the current location as initial state.
- 2) Set delivery location as goal state.
- 3) Generate all possible routes.
- 4) Calculate $f(n) = g(n) + h(n)$.
- 5) Select the route with minimum $f(n)$.
- 6) move the drone along the selected route.

Result.

The goal-based agent with A Search helps the drone choose a safe, shorter and cost effective route. it can also change its route when obstacles are detected.

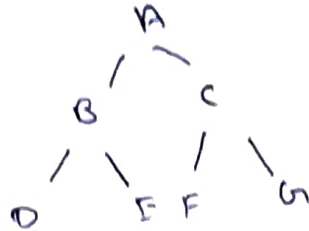
2) Iterative Deepening ~~Search~~ Search (IDS)

Probe understanding.

Iterative Deepening Search (IDS) combines the advantages of Depth-First Search (DFS) and Breadth-First Search (BFS). it repeatedly perform depth-limited search by increasing the depth limit.

State, consider the graph.

Plain tree.



Let A be the starting node and F be the goal node.

method Selection.

IDS begins with depth limit 0 and increases the limit until the goal is found.

Solution process.

Depth Limit 0:

A
Goal F is not found.

Depth Limit 1:

$A \rightarrow B \rightarrow C$

Goal is still not found.

Depth Limit 2:

$A \rightarrow B \rightarrow D \rightarrow E \rightarrow C \rightarrow F$

Now the goal F is found.

Therefore, the solution path is:

$A \rightarrow C \rightarrow F$

Accuracy and Result.

the time complexity is.

$O(b^d)$

the space complexity is:

$O(bd)$.

IDS is complete like BFS because it searches each depth level, while it is space efficient like DFS because it does not store all nodes in memory.

3) Missionaries and cannibals using BFS.

Problem Understanding

There are 3 missionaries and 3 cannibals on one side of a river. They must cross using a boat that can carry a maximum of 2 people.

The main condition is that cannibals³ must never outnumber missionaries on either bank when missionaries are present.

Problem Formulation.

Represent each state as:
 (M, C, B)

where

- * M = missionaries on the left bank
- * C = cannibals on the left bank.
- * B = boat position.

Initial State.

$(3, 3, L)$

Goal State.:

$(0, 0, R)$

Possible Actions.

The boat can carry

- * 2 missionaries.
- * 2 cannibals.
- * 1 missionary + 1 cannibal.
- * 1 missionary.

After every movement, the safety condition must be checked.

First three State using BFS

State 1 - initial State.

$(3, 3, L)$

two cannibals cross the river.

$(3, 3, L) \rightarrow (3, 1, R)$

State 2:

$(3, 1, R)$

one cannibal returns:

$(3, 1, R) \rightarrow (3, 2, L)$

State 3:

$(3, 2, L)$

therefore, one valid beginning is

$(3, 3, L) \rightarrow (3, 1, R) \rightarrow (3, 2, L)$

Result.

~~Breadth - First Search (BFS)~~ explores state level by level and rejects unsafe state since each boat crossing has equal cost, BFS and find a solution with the minimum num.